An explainable algorithm for credit risk prediction based on contrast patterns

Erwis Melchor Pérez[1 [0000−1111−2222−3333]], José Francisco Martínez-Trinidad[1 [1111−2222−3333−4444]], and Jesús Ariel Carrasco Ochoa[1 [2222−−3333−4444−5555]]

Instituto Nacional de Astrofísica, Óptica y Electrónica
{emelchor,fmartine,ariel}@inaoep.mx

Abstract. Resumen

Keywords: credit risk · pattern recognition · clustering.

1 Introduction

Credit risk prediction involves determining whether an applicant will default on a future financial obligation [1], and is therefore a fundamental task in the financial sector. Credit risk prediction is commonly considered as a supervised classification task; the goal is to assign each applicant to a class (e.g., default or non-default), rather than estimating a continuous predictive risk value. Credit risk prediction directly impacts the stability of banks and financial institutions, where accurate and explainable predictions are necessary to ensure regulatory compliance [2, 3]. Increasingly, financial institutions have implemented evaluations and management processes that employ pattern recognition and machine learning algorithms for credit risk prediction [4, 5]. As these predictors become more widely adopted, improving default prediction enables more accurate credit risk prediction and reduces losses [6–8].

Recent research in credit risk prediction has prioritized improving the performance of credit risk prediction algorithms based on machine learning and pattern recognition through the use of oversampling or undersampling methods, thereby reducing algorithmic bias towards the majority class [9–13]; the use of feature selection methods for identifying the most relevant features of a dataset to improve the performance of prediction algorithms [1, 14–16]; and the use of explainable AI techniques [10, 17–19] that allow users to comprehend and trust the predictions provided by credit risk prediction algorithms, which makes the predictions of the algorithm to comply with regulations.

Machine learning algorithms have demonstrated good predictive performance, and decision trees are among the most effective for credit risk prediction [4, 20–25]. However, the need for explanations remains a significant limitation, particularly in the financial sector, as financial institutions and regulators require justification for their decisions [26, 27]. Several authors have addressed the lack of explainability in credit risk prediction [17, 28, 29].

Explainability techniques, also known as XAI (eXplainable Artificial Intelligence), aim to make machine learning prediction processes transparent and understandable to humans. These techniques are important in areas such as credit risk prediction, where justifying and explaining the predictions is essential in enabling financial institutions to make more informed decisions. Several recent studies have applied SHAP [30] and LIME [31] to credit risk prediction to generate explanations for individual predictions [32–34]. SHAP assigns to each feature a value that represents how much it contributed to the prediction. These values help explain the role each feature played in the prediction. LIME builds a simple interpretable predictor that approximates the behaviour of the original predictor by modifying a specific instance and observing the changes in the prediction. The result is an explanation that highlights the features contributing most to the prediction, showing how each affects the prediction in a positive or negative direction. However, these techniques focus on individual features and often overlook how several features jointly contribute to the prediction.

In credit risk prediction, a predictor's output depends on multiple features rather than individual features. Therefore, explainable predictors should consider multiple features to explain the prediction [18, 35]. In response to this need, this paper proposes a credit risk prediction algorithm that generates explanations based on contrast patterns, which are conjunctions of conditions of the form $feature\#value$ that are frequent in a class but rarely appear in the opposite class. Since contrast patterns can be extracted from decision-tree-based, we propose using decision-tree-based predictors to predict credit risk, which have been reported to perform competitively compared to the most successful approaches. In this way, the proposed algorithm leverages the affinity and synergy between decision-tree-based predictors, and contrast patterns extracted from decision treed.

Unlike other explainable algorithms that merely highlight the importance of individual features, our method explains predictions based on contrast patterns present in the training data and shared with the new applicant. Experiments conducted on both public and private financial datasets show that the proposed algorithm achieves competitive predictive accuracy while offering instance-level explanations based on the contrast patterns shared with other clients in the class, unlike the current state-of-the-art, which explains based on the single features contributing most to the prediction.

The remainder of this paper is organized as follows. Section 2 presents the related work. Section 3 describes the proposed algorithm. Section 4 presents our experimental results and a discussion about them. Finally, Section 5 presents our conclusions and future work directions.

2 Related works

Credit risk prediction has been extensively studied in the literature using a wide range of pattern recognition and machine learning algorithms [3, 7, 8, 10, 14, 15, 28, 29, 33, 36–48]. In this section, we review the most recent works on credit risk prediction, with a focus on approaches that either prioritize achieving the highest accurate credit risk prediction or incorporate explainability for credit risk prediction. The works considered in this review are those that utilize publicly available datasets, commonly used in the literature and accessible online, which allows a comparison against our proposal. Accordingly, this section is divided into two parts: works that focus solely on accuracy without offering explanations, and works that integrate explainability into their predictors.

2.1 Predictors for Credit Risk Prediction

Recent works on credit risk prediction have focused on developing predictive algorithms to improve performance metrics, such as accuracy, area under the curve (AUC), Type II error, or F1-score.

In [38], the authors proposed a stacked ensemble for credit risk prediction that combines Random Forests (RF), Logistic Regression (LR), and Convolutional Neural Networks (CNN). The results of these predictors are used as input features for a Multilayer Perceptron (MLP), which learns to combine them to produce the prediction. The authors used SMOTEENN [49] to address class imbalance, and the stacked ensemble is compared against CNN, LR, MLP, and RF. After applying SMOTE-ENN to address class imbalance, the stacked ensemble consistently outperformed the individual predictors, achieving accuracies near 96%.

A comparative study of different predictors for credit risk prediction is presented in [46], including LR, RF, and Artificial Neural Networks (ANN). In this study, SMOTE [50] was employed to balance the training dataset, leading to significant improvements in the performance metrics of all predictors. Among them, RF achieved the best results, with an accuracy of 98%.

In [47], Deep Convolutional Neural Networks (DCNN) and Support Vector Regression (SVR) were compared for credit risk prediction. The application of SMOTE contributed to improved predictive performance in both predictors. Using SMOTE to balance the dataset, SVR achieved an accuracy of 98% and F1-score of 87%.

In [51], the authors present the Level-Wise Feature-Guided Cascading Ensembles (LFGCE) model, which integrates hierarchical feature selection with a cascading ensemble learning scheme. To evaluate its performance, several predictors were employed, including RF, Support Vector Machines (SVM), LR, Decision Trees (DT), ANN, Linear Discriminant Analysis (LDA), and XGBoost.

Among these, XGBoost achieved the best results in both feature selection and prediction, with an AUC of 94%. However, it is important to note that the study does not address class imbalance, as no data balancing methods were applied.

Most of the reviewed studies emphasise improving predictive performance through class balancing methods to mitigate data imbalance. Several predictors yield good results in credit risk prediction. However, the reported performance metrics (such as accuracy, AUC, and F1-score) vary significantly on the chosen predictors and balancing methods. Overall, the authors consistently point out that addressing class imbalance is crucial for achieving reliable predictions, with many studies reporting high accuracy values, often exceeding 90%.

2.2 Explainable Predictors for Credit Risk Prediction

Incorporating explainability into credit risk prediction has been a growing area of study. These studies aim to achieve robust predictive performance while prioritising the explainability of predictions.

A study by [43] addresses credit risk prediction by comparing the predictive performance of XGBoost, RF, Least Squares SVM (LSSVM), and a CNN. After applying SMOTE to balance the dataset, XGBoost achieved the best predictive accuracy, reaching 92%. To enhance predictor explainability, SHAP analysis was applied to quantify the individual contributions of each feature. From the predictors evaluated, XGBoost performed best in terms of predictive accuracy and also provided explanations through SHAP values.

In [33], the authors present an ensemble framework that combines RF and XGBoost using probability-weighted averaging. The framework was evaluated using eight predictors, including LR, DT, Gradient Boosting (GB), LightGBM (LGBM), MLP, Naive Bayes (NB), and XGBoost. Their proposed ensemble significantly outperformed all others, achieving an AUC of 97%. SHAP was employed to explain the predictions, which allowed them to determine the contribution of each feature.

In [29], XGBoost is used as a predictor, combined with SHAP, LIME, and K-means to enhance explainability, achieving an AUC of 78%. SHAP and LIME are used to generate feature-contribution vectors for each training instance, which are grouped using K-means. For explaining the prediction of a new instance, its feature-contribution vector is calculated and assigned to the nearest cluster, and the prediction explanation is provided as the cluster's average vector.

In [3], the authors present a comparative analysis of the performance of boosting algorithms, including LightGBM, XGBoost, CatBoost, and AdaBoost. Additionally, SMOTE-Tomek [52] is employed to balance the dataset. The results indicate that LightGBM achieves the best performance with 95% accuracy and 98% AUC, outperforming the other boosting methods. SHAP is applied to quantify the contribution of each feature to the prediction.

In [28], a multilayer dynamic graph neural network (DYMGNN), proposed for credit risk prediction. The authors compare their approach against several baselines: LR, XGBoost, and DNN. Unlike many related studies, no resampling or class balancing methods are applied. Experimental results show that the DYMGNN achieves the best results with 81% accuracy and 83% F1-score. For explainability, SHAP is used to identify the features that contribute most to the predictions.

In [48], the authors propose an explainable two-stage technique to explain the results of credit risk prediction. In the first stage, SHAP is applied to analyse the importance of features according to their positive or negative impact on the prediction. In the second stage, they use a genetic algorithm to generate counterfactual explanations by perturbing the features according to this categorisation, thereby identifying the minimum necessary changes to modify the prediction. This technique was evaluated using multiple predictors, including SVM, XGBoost, MLP, and LSTM, with an accuracy of 90% for the employed predictors. The results demonstrate that the features modified most frequently closely align with the features identified as most relevant by SHAP.

In [53], the authors compare the performance of several Deep Learning (DL) architectures for credit risk prediction, including Convolutional Neural Networks (CNN), Long Short-Term Memory (LSTM) networks, Gated Recurrent Units (GRU), and Graph Neural Networks (GNN). To address class imbalance, they apply SMOTEENN. The experimental results show that GRU outperformed all other predictors, achieving an accuracy of 91%. To explain the prediction, SHAP is used to quantify the importance of features and their contributions to the predictions.

In [7], the authors propose an approach for credit risk prediction on imbalanced datasets by integrating the TabTransformer model with a weighted loss function. This approach helps mitigate the bias toward the majority class, improving predictive performance on minority classes, achieving 91% accuracy and an AUC of 93%. SHAP is employed to evaluate feature importance and identify those features that contribute most to the predictions.

In [54], the authors propose a robust framework for credit risk prediction on imbalanced datasets by introducing the K-SMOTEENN method, which integrates oversampling and noise-filtering methods with a stacking ensemble. The ensemble combines Random Forest, XGBoost, and LightGBM, resulting in a predictive performance of 92% F1-score, and 96% AUC. SHAP is used to analyse and explain the contributions of individual features to the prediction.

As discussed in the above works, credit risk prediction has been addressed through a wide range of predictors, balancing methods, feature selection methods, and evaluated through multiple performance metrics. Nevertheless, accurately detecting defaults remains a significant challenge, as no single method has consistently outperformed others across different scenarios. Moreover, the most commonly used explainability approaches, such as SHAP and LIME, while helpful in assessing the individual contribution of features, fail to capture feature value combinations that characterise different types of applicants. This limitation is particularly relevant, since financial institutions increasingly demand not only high predictive accuracy but also clear and case-specific justifications, both to meet regulatory requirements and to improve communication with applicants.

3 Proposed explainable algorithm

As discussed in the previous section, recent research has increasingly focused on explaining credit risk predictions. Techniques such as SHAP and LIME, commonly used for explaining credit risk predictions, assign importance scores to individual features; however, they do not capture the joint effect of multiple features in a prediction, which is particularly relevant in credit risk prediction. This limitation highlights the need for approaches that go beyond individual feature contributions and consider multiple features when explaining credit risk predictions.

The idea behind the proposed Pattern-based Credit Risk Prediction and Explanation (PbCRPE) algorithm is to enable both credit risk prediction and its explanation to rely on the same approach. Unlike traditional explainers for credit risk prediction, which apply a predictor that follows an approach different from the one used for explaining predictions (e.g., SHAP or LIME which are commonly based on individual feature contribution techniques), PbCRPE predicts and explains following a similar approach.

As mentioned previously, credit risk prediction depends on multiple features rather than individual features. For instance, a 'high debt-to-income ratio' may only indicate default when combined with 'low job stability'. This observation highlights the need for explanation approaches that consider multiple features. In this context, PbCRPE addresses this need by using contrast patterns to explain predictions. Contrast patterns allow providing explanations based on conjunctions of $feature\#value$ conditions, similar to those used by decision trees. This structural similarity motivates the use of a predictor based on decision trees, not only because of their alignment with contrast patterns, but also because of their strong performance in state-of-the-art credit risk prediction [38]. A decision tree can be interpreted as a hierarchy of decision rules, where each path from the root to a leaf node corresponds to a conjunction of conditions of the form $feature\#value$. Similarly, contrast patterns are expressed as conjunctions of conditions of the form $feature\#value$. Based on this relationship, the proposed algorithm is organized into two main stages: (i) a training stage, which includes predictor training and contrast pattern mining from the classes, and (ii) a prediction and explanation stage using the trained predictor and the mined contrast patterns.

3.1 Training and contrast pattern mining stage

The first stage of the PbCRPE consists of training a classifier for a credit risk predictor and mining contrast patterns, which are expressed as a conjunction of $feature\#value$ conditions that will later be used to explain the classifier's predictions.

For credit risk prediction, we propose using the FT4cip predictor [55], a decision tree-based predictor capable of handling mixed and imbalanced datasets, which has reported strong predictive quality on such data, making it suitable for this task. Although FT4cip is designed to handle imbalanced datasets, the experiments conducted in this work indicated that it achieved better predictive quality when trained using a previously balanced training set. Thus, following the results reported in [56], we propose applying SMOTENC [50, 57] to the training set to balance it prior to predictor training.

For mining contrast patterns, since instances belonging to the same class do not necessarily share the same $feature\#value$ groups of similar instances are first identified through clustering. Contrast patterns are subsequently mined independently for each cluster with respect to the whole opposite class, yielding contrast patterns that characterize instances in the class specific to both the cluster and its associated class. To generate these clusters, we propose using the Kmodes algorithm [58, 59], as it is specifically designed for mixed data and avoids the limitations of distance-based methods when handling non-numeric features. To determine the number of clusters, following [60, 61], we propose using the Silhouette coefficient, which, according to these works, enables the identification of well-defined clusters.

For mining contrast patterns, we propose using CPMining, introduced in the PBC4cip algorithm [62, 63], as a supervised contrast pattern miner that explicitly exploits class differences to identify contrast patterns. CPMining is applied independently to each cluster by contrasting the instances of the cluster with all instances of the opposite class. This design allows mining contrast patterns that represent $feature\#value$ conditions observed among similar instances within each cluster while appearing less in instances from the opposite class.

Due to the large number of contrast patterns that can be generated during mining, the filtering used in CPMining is applied to reduce redundant patterns. This filtering removes more specific patterns when a more general pattern covers the same instances, and when redundant conditions are present within a pattern.

Additionally, we propose removing contrast patterns whose support in the opposite class is non-zero. As a result, the final set contains only contrast patterns whose support in the opposite class is zero.

3.2 Prediction and explanation stage

Given a new instance x, the prediction and explanation stage first determines its class (defaulter or non-defaulter) using the trained FT4cip predictor. Once the predicted class c for x has been determined, the contrast patterns mined from each cluster associated with class c are used to compute a similarity score between the new instance x and each cluster. The cluster with the highest similarity score is selected as the cluster most similar to the new instance.

To identify the cluster that best represents a new instance x, PbCRPE defines a contrast pattern-based similarity score between x and each cluster $C_{c,j}$ associated with the predicted class c. Unlike conventional similarity measures for mixed data, which compare instances directly using their feature values, the proposed score evaluates similarity through the contrast patterns mined from each cluster. These patterns summarize the characteristic combinations of $feature\#value$ conditions shared by similar instances within the cluster and associated with the predicted class, making them a more appropriate representation for explanation purposes. Consequently, the similarity between x and a cluster is quantified according to the support accumulated by the cluster's contrast patterns that cover the new instance. Specifically, the similarity score is computed as the ratio between the accumulated support of the covering contrast patterns and the total support of all contrast patterns mined from the cluster, as defined in Equation 1.

$$Sim(x, C_{c,j}) = \frac{\sum\limits_{p \in \mathcal{P}_{c,j}(x)} supp(p)}{\sum\limits_{p \in \mathcal{P}_{c,j}} supp(p)}, \tag{1}$$

This similarity score increases as the support of the contrast patterns covering the new instance increases. Since contrast patterns with higher support cover more similar instances within the cluster, they contribute more to the similarity score. Furthermore, dividing by the total support of the contrast patterns mined from the cluster avoids bias towards clusters with more contrast patterns. Consequently, higher similarity scores indicate that a larger proportion of the support of the contrast patterns mined from the cluster is associated with to contrast patterns that cover the new instance.

The cluster with the highest similarity score is selected as the cluster most similar to the new instance x. If multiple clusters obtain the same highest similarity score, the cluster whose highest-support contrast pattern covering the new instance has the greatest support is selected. If a tie persists, the cluster whose corresponding highest-support contrast pattern contains the largest number of $feature\#value$ conditions is selected. This tie-breaking criterion favors the cluster whose covering contrast pattern has the greatest support and, when support is equal, provides the most specific combination of conditions. The explanation stage assumes that at least one contrast pattern mined from the clusters associated with the predicted class covers the new instance. Under this assumption, empirically validated later a Section 4the most similar cluster always contains at least one covering contrast pattern. Once the most similar cluster C_{c,j^*} has been identified, the contrast pattern p^* with the highest support is selected from $\mathcal{P}_{c,j^*}(x)$, the set of contrast patterns from C_{c,j^*} that cover the instance x, as defined in Equation 2. Support is used as the selection criterion because contrast patterns with higher support cover a larger number of similar instances within the selected cluster, indicating that more instances share the same $feature\#value$ conditions.

$$p^* = \arg \max_{p \in \mathcal{P}_{c,j^*}(x)} supp(p), \tag{2}$$

The selected contrast pattern becomes the final explanation of the prediction and is expressed as a conjunction of $feature\#value$ conditions. Since the selected contrast pattern covers multiple similar instances within the most similar cluster, the explanation is based on conditions shared by the new instance and those covered instances. If multiple patterns achieve the same maximum support, the pattern containing the largest number of $feature\#value$ conditions is selected, since it provides a more detailed explanation by incorporating a larger set of conditions associated with the covered instances. If multiple contrast patterns remain tied after applying this criterion, all tied contrast patterns will be the explanation.

Unlike recent works, where explainability highlights individual features that contribute most to the prediction, showing how each affects the prediction in a positive or negative direction. In our proposal, explainability involves several features that jointly contribute to the prediction.

4 Experimental Results

The objective of this experimental evaluation is to assess whether PbCRPE can achieve competitive credit risk prediction quality compared with other state-of-the-art works while simultaneously providing explainable predictions based on combinations of multiple features that jointly contribute to the prediction. The evaluation considers two complementary aspects of prediction quality, assessed using AUC and Type II Error, two commonly used metrics in credit risk prediction that capture different aspects of predictive quality: the overall ability of the predictor to discriminate between defaulters and non-defaulters and its ability to correctly identify defaulters, respectively. While a competitive AUC indicates that PbCRPE maintains a strong overall discrimination capability, particular emphasis is placed on Type II Error because incorrectly classifying a defaulter as a non-defaulter may result in direct financial losses for lending institutions. Therefore, achieving a low Type II Error is particularly relevant from a practical credit risk perspective, and the experimental evaluation investigates whether PbCRPE can provide an advantage in this aspect while maintaining competitive overall prediction quality.

PbCRPE is compared with twelve recent state-of-the-art credit risk prediction works using the same experimental setup and twelve public and private credit risk datasets. These works were selected from studies published between 2024 and 2025 because they represent recent developments in machine-learning-based and explainable credit risk prediction, report competitive prediction results, and evaluate their algorithms on benchmark datasets that overlap with or are comparable to those considered in this work. The results show that PbCRPE achieves competitive prediction quality in terms of AUC compared with the evaluated state-of-the-art works, while obtaining the best overall result in terms of Type II Error. This result is particularly relevant from a practical perspective because a lower Type II Error indicates a lower proportion of actual defaulters incorrectly classified as non-defaulters. Therefore, the experimental evaluation shows that PbCRPE maintains competitive overall prediction quality while achieving a particular advantage in identifying actual defaulters and reducing the risk of incorrectly classifying them as non-defaulters.

4.1 Datasets

The experimental evaluation was conducted using twelve credit risk datasets obtained from both public and private sources. The datasets were selected because they represent diverse credit risk scenarios while sharing the two characteristics addressed by the proposed PbCRPE algorithm: mixed-type data, including numerical and categorical features, and different degrees of class imbalance. The diversity in feature types and class distributions allows the prediction quality of PbCRPE to be evaluated under different data conditions.

Table 1 summarizes the main characteristics of the datasets, including the number of instances, number of features (total, numerical, and categorical), class distribution, imbalance ratio (IR), and data source. The eleven public datasets were obtained from the UCI Machine Learning Repository[1] and Kaggle[2], two widely used repositories for benchmarking credit risk prediction algorithms. The twelfth dataset was provided by a private financial institution. Due to a confidentiality agreement, the identity of the institution and further information about the dataset cannot be disclosed. The datasets vary substantially in both size and class distribution, ranging from 628 to 150,000 instances and from nearly balanced datasets (IR = 1.09) to highly imbalanced datasets (IR = 13.96). Consequently, the experimental evaluation covers a broad spectrum of credit risk

[1] `https://archive.ics.uci.edu/`
[2] `https://www.kaggle.com/`

scenarios, allowing the prediction quality of PbCRPE to be assessed under different levels of class imbalance.

Table 1: Characteristics of the twelve credit risk datasets used in the experimental evaluation.

Dataset	Instances	Features	Numerical/Categorical	No-defaulter	Defaulter	IR	Domain
AER	1,319	12	9/3	1,023	296	3.45	Public
Australia	690	14	6/8	307	383	1.24	Public
Japan	690	15	6/9	307	383	1.24	Public
EIZ	5,510	22	13/9	2,295	3,215	1.40	Private
German	1,000	20	7/13	700	300	2.33	Public
HMEQ-Data	3,364	12	10/2	3,064	300	10.21	Public
Taiwan	30,000	24	20/3	23,364	6,636	3.52	Public
GMSC	150,000	11	10/1	139,974	10,026	13.96	Public
Heloc	10,459	23	23/0	5,000	5,459	1.09	Public
Loan data set	628	12	7/4	332	296	1.12	Public
PPDai	55,596	29	22/7	48,413	7,183	6.74	Public
Thomas dataset	1,225	14	14/0	902	323	2.79	Public

4.2 Credit risk prediction quality

Figures 1 and 2 compare the prediction quality of PbCRPE with that of the twelve selected state-of-the-art approaches using AUC and Type II Error, respectively. For each dataset, a stratified 5-fold cross-validation procedure was employed, in which the data were divided into five folds while preserving the class distribution across folds. In each iteration, four folds (approximately 80% of the instances) were used for training and the remaining fold (approximately 20%) was used for testing. The process was repeated five times, with each fold serving once as the test set. The same cross-validation protocol was applied to PbCRPE and all compared approaches to ensure a fair and consistent evaluation. The reported AUC and Type II Error values correspond to the mean results obtained across the five folds. Figure 1 reports the mean AUC across the twelve evaluated datasets, whereas Figure 2 presents the corresponding mean Type II Error across the same datasets.

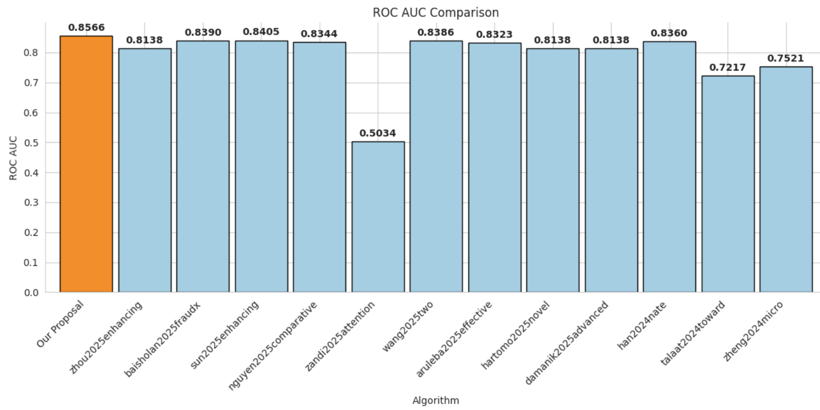

Fig. 1: Mean AUC obtained by PbCRPE and the compared state-of-the-art approaches across the twelve evaluated credit risk datasets. Higher values indicate better prediction quality.

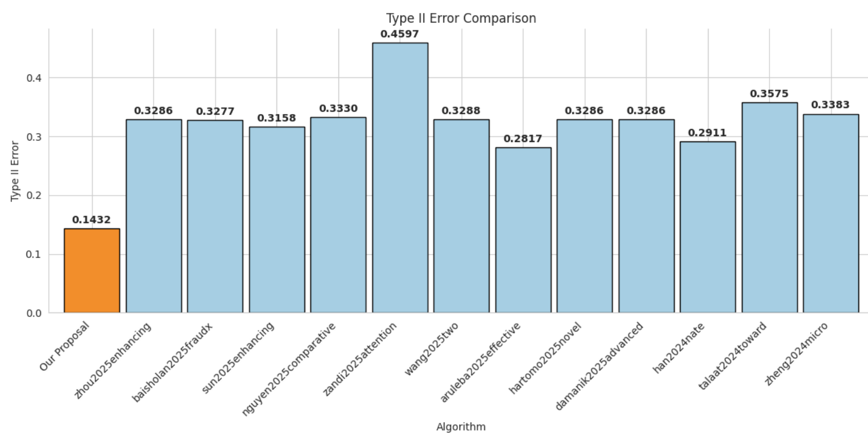

Fig. 2: Mean Type II Error obtained by PbCRPE and the compared state-of-the-art approaches across the twelve evaluated credit risk datasets. Lower values indicate better prediction quality.

Across the twelve evaluated datasets, PbCRPE achieves a mean AUC of **0.8566** and a mean Type II Error of **0.1432**. The AUC results indicate that PbCRPE provides competitive prediction quality compared with the selected state-of-the-art approaches. In particular, PbCRPE attains the highest AUC on several datasets while maintaining competitive results on the remaining datasets. In terms of Type II Error, PbCRPE achieves the best overall result among the evaluated approaches, indicating a lower proportion of actual defaulters incorrectly classified as non-defaulters.

To further assess the statistical significance of the observed differences, the Friedman test was applied separately to the AUC and Type II Error results across the twelve evaluated datasets. This non-parametric test evaluates whether statistically significant differences exist among the compared approaches based on their ranks across datasets. For both metrics, the best-performing approach within each dataset was assigned rank 1, with higher rank values corresponding to lower prediction quality. Consequently, lower average ranks indicate better overall prediction quality across the evaluated datasets. The corresponding Friedman test results are presented in Figures 3 and 4.

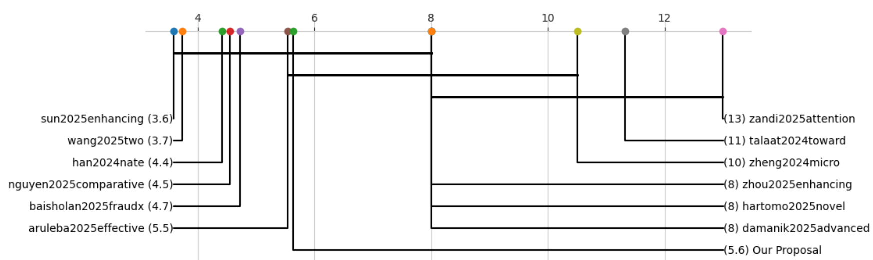

Fig. 3: Friedman test results for AUC across the twelve evaluated credit risk datasets.

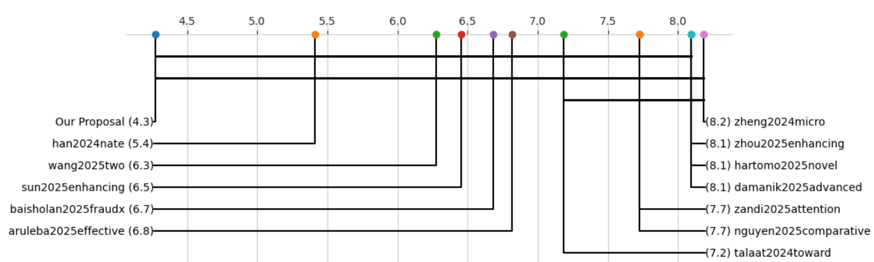

Fig. 4: Friedman test results for Type II Error across the twelve evaluated credit risk datasets.

For AUC, the statistical analysis indicates that PbCRPE remains competitive with the state-of-the-art works, as reflected by its relative ranking across the evaluated datasets. Although PbCRPE does not necessarily outperform all compared approaches in terms of AUC, its overall ranking demonstrates that its prediction quality is comparable to that of recent works reported in the literature.

While the AUC results show that PbCRPE provides competitive prediction quality compared with the evaluated works, the Type II Error results show that PbCRPE achieves the best overall result among the evaluated works. The proposed algorithm obtains the lowest mean Type II Error and the best mean rank for this metric across the evaluated datasets. This result is particularly relevant from a practical credit risk perspective because Type II Error measures the proportion of actual defaulters incorrectly classified as non-defaulters. A lower value therefore indicates a greater ability to identify customers who are likely to default, reducing the risk of incorrectly considering them as non-defaulters.

To determine which pairwise differences between works are statistically significant, a Nemenyi post-hoc test was performed following the Friedman analysis. The resulting comparisons provide additional evidence regarding the relative position of PbCRPE with respect to the evaluated state-of-the-art works. For AUC, the post-hoc analysis shows that PbCRPE remains within the group of competitive works, with no statistically significant disadvantage with respect to the evaluated works. For Type II Error, PbCRPE obtains the lowest mean rank, consistent with its lowest mean Type II Error across the evaluated datasets. These results indicate that the proposed algorithm achieves the best overall position for this metric while maintaining competitive AUC results.

From a practical perspective, the results obtained for Type II Error are particularly important in credit risk prediction because failing to identify a defaulter may result in direct financial losses for lending institutions. Therefore, the low Type II Error obtained by PbCRPE reflects its ability to reduce the proportion of actual defaulters incorrectly classified as non-defaulters. This result is particularly relevant because PbCRPE achieves the lowest Type II Error while maintaining competitive overall discrimination capability in terms of AUC. Taken together, these results indicate that PbCRPE provides competitive prediction quality compared with recent state-of-the-art works while achieving the best overall results in terms of Type II Error.

4.3 Discussion

PbCRPE achieves competitive prediction quality compared with the twelve evaluated state-of-the-art works while simultaneously providing an explanation for each prediction. This dual capability, integrating prediction and explanation within a single algorithm, represents an important advantage over post-hoc explanation techniques such as SHAP and LIME, which typically express explanations through feature-level contribution or importance scores rather than explicitly representing a conjunction of feature-value conditions. This distinction is particularly relevant in credit risk prediction, where default risk is generally associated with combinations of financial and personal attributes rather than individual features considered in isolation [38].

A key contribution of PbCRPE is therefore the representation of explanations as contrast patterns rather than as independent feature-importance scores. While SHAP assigns a contribution value to each feature according to its contribution to the prediction, and LIME estimates the local importance of features using an interpretable surrogate model, PbCRPE provides an explicit conjunction of $feature\#value$ conditions that jointly characterizes the prediction. Consequently, the explanation not only identifies which features are involved in the prediction, but also specifies the particular combination of conditions shared by the new instance and multiple similar instances within the selected cluster. For example, a pattern such as $credit\_history \neq \texttt{A34} \land amount < \$7{,}824 \land duration > 28\,\text{months} \land installment\_rate > 2\% \land num\_loans \leq 1$ represents a joint configuration of conditions rather than a set of independent feature contributions. This representation preserves the relationships among the conditions and provides an explanation associated with a group of similar instances rather than with isolated feature contributions.

This difference should not be interpreted as indicating that SHAP or LIME cannot account for feature interactions. Rather, the distinction lies in how the resulting explanation is represented. SHAP and LIME primarily summarize the local contribution or importance of individual features, whereas PbCRPE explicitly expresses the explanation as a conjunction of conditions that jointly cover the instance and are shared by similar instances. Therefore, PbCRPE provides a pattern-based representation that can be directly interpreted as a combination of $feature\#value$ conditions associated with the predicted class.

To illustrate the explanation process, Table 2 presents an example of an instance from the German credit dataset that was classified by PbCRPE as *defaulter*.

Table 2: Example instance from the German credit dataset.

Feature	Type	Value	Description
estatus_cuenta	C	A14	No checking account
duracion	N	36	Duration of the credit: 36 months
historial_creditos	C	A30	No credits taken/all credits paid back duly
finalidad	C	A43	Radio/television
monto_credito	N	3342	Credit amount: $3,342
cuenta_ahorro	C	A65	Unknown/no savings account
empleo_actual	C	A75	Present employment for 7 years or more
tasa_interes	N	4	Installment rate: 4% of disposable income
genero_estadocivil	C	A93	Male, single
aval	C	A101	No other debtors or guarantors
residencia	N	2	Present residence since 2 years
propiedad	C	A123	Car or other property, excluding real estate and savings agreements/life insurance
edad	N	51	Age: 51 years
planesdepago	C	A143	No other installment plans
vivienda	C	A152	Own housing
numero_creditos	N	1	One existing credit at this bank
empleo	C	A173	Skilled employee/official
dependientes	N	1	One person liable to receive maintenance
telefono	C	A192	Telephone registered under the customer's name
trabajo_extranjero	C	A201	Foreign worker: yes

For the instance shown in Table 2, PbCRPE first considers the clusters associated with the predicted class. Using the contrast pattern-based similarity score defined in Equation 1, the cluster with the highest similarity to the new instance is selected. Subsequently, among the contrast patterns from the selected cluster that cover the new instance, the pattern with the highest support is selected according to the criterion defined in Equation 2:

$$credit\_history \neq \texttt{A34} \wedge amount < \$7{,}824 \wedge duration > 28\,\text{months}$$
$$\wedge\ installment\_rate > 2\% \wedge num\_loans \leq 1$$

Thus, the selected contrast pattern covers the instance because all of its $feature\#value$ conditions are simultaneously satisfied. For example, the condition $credit\_history \neq \texttt{A34}$ is satisfied because the instance has an *No credits taken/all credits paid back duly* credit history, which corresponds to a value different from `A34`. Similarly, $amount < \$7{,}824$ is satisfied because the credit amount is \$3,342; $duration > 28\,\text{months}$ is satisfied because the duration is 36 months; $installment\_rate > 2\%$ is satisfied because the installment rate is 4%; and $num\_loans \leq 1$ is satisfied because the instance has one existing loan. Since a contrast pattern is defined as a conjunction of conditions, all these conditions must be satisfied simultaneously for the pattern to cover the instance. Importantly, the pattern is not interpreted as a collection of independent feature contributions. Rather, its conjunction represents a combination of $feature\#value$ conditions shared by the new instance and multiple similar instances within the selected cluster. Consequently, the contrast pattern provides a concise representation of the combination of conditions associated with the predicted credit risk class.

The example also illustrates the role of performing contrast pattern mining at the cluster level rather than directly at the class level. Instead of extracting patterns intended to represent all instances belonging to the same class, PbCRPE mines contrast patterns from groups of similar instances. This strategy allows different local structures within the same class to be captured, enabling the resulting patterns to represent $feature\#value$ conditions that are more specific to particular groups of similar instances. Therefore, the resulting explanations are associated not only with the predicted class but also with a specific group of similar instances represented by the selected cluster.

The experimental results confirm that the coverage condition required by the explanation procedure was satisfied for all evaluated instances. In particular, for every evaluated instance across all twelve datasets, at least one contrast pattern mined from a cluster associated with the predicted class covered the instance. Therefore, the similarity score could be computed from covering contrast patterns for all evaluated instances, and the explanation procedure could be applied without requiring any fallback mechanism. These results provide empirical evidence that the coverage condition defined in the explanation procedure was consistently satisfied under the experimental conditions considered in this study.

Taken together, these results indicate that PbCRPE combines competitive prediction quality with explanations based on contrast patterns that represent combinations of $feature\#value$ conditions shared by similar instances. By integrating prediction for mixed-type and imbalanced data with cluster-level contrast pattern mining and representative pattern selection, PbCRPE generates explanations that are directly associated with groups of similar instances rather than isolated feature contributions. These results support the applicability of the proposed algorithm to credit risk datasets involving mixed-type features and class imbalance, where both prediction quality and explainable decision support are relevant.

References

[1] Jianxin Zhu, Xiong Wu, Lean Yu, and Xiaoming Zhang. "A hybrid clustering and boosting tree feature selection (CBTFS) method for credit risk

assessment with high-dimensionality". In: *Technological and Economic Development of Economy* (2025), pp. 1–33. DOI: 10.3846/tede.2025.23060.

[2] Xiaosong Zhao, Yong Liu, and Qiangfu Zhao. "Improved LightGBM for extremely imbalanced data and application to credit card fraud detection". In: *IEEE Access* (2024). DOI: 10.1109/ACCESS.2024.3487212.

[3] Nhat Nguyen and Duy Ngo. "Comparative analysis of boosting algorithms for predicting personal default". In: *Cogent Economics & Finance* 13.1 (2025), p. 2465971. DOI: 10.1080/23322039.2025.2465971.

[4] Patrick Weber, K Valerie Carl, and Oliver Hinz. "Applications of explainable artificial intelligence in finance—a systematic review of finance, information systems, and computer science literature". In: *Management Review Quarterly* 74.2 (2024), pp. 867–907. DOI: 10.1007/s11301-023-00320-0.

[5] Wei Jie Yeo, Wihan Van Der Heever, Rui Mao, Erik Cambria, Ranjan Satapathy, and Gianmarco Mengaldo. "A comprehensive review on financial explainable AI". In: *Artificial Intelligence Review* 58.6 (2025), pp. 1–49. DOI: 10.1007/s10462-024-11077-7.

[6] Sichong Lu, Yi Su, Xiaoming Zhang, Jiahui Chai, and Lean Yu. "LLM-infused bi-level semantic enhancement for corporate credit risk prediction". In: *Information Processing & Management* 62.4 (2025), p. 104091. DOI: 10.1016/j.ipm.2025.104091.

[7] Kristoko Dwi Hartomo, Christian Arthur, and Yessica Nataliani. "A Novel Weighted Loss TabTransformer Integrating Explainable AI for Imbalanced Credit Risk Datasets". In: *IEEE Access* (2025). DOI: 10.1109/ACCESS.2025.3541878.

[8] Tao Yu, Wei Huang, Xin Tang, and Duosi Zheng. "A hybrid unsupervised machine learning model with spectral clustering and semi-supervised support vector machine for credit risk assessment". In: *PloS one* 20.1 (2025), e0316557. DOI: 10.1371/journal.pone.0316557.

[9] I Nyoman Mahayasa Adiputra, Pei-Chun Lin, and Paweena Wanchai. "The Effectiveness of Generative Adversarial Network-Based Oversampling Methods for Imbalanced Multi-Class Credit Score Classification". In: *Electronics* 14.4 (2025), p. 697. DOI: 10.3390/electronics14040697.

[10] Shaukat Ali Shahee and Rujavi Patel. "An Explainable ADASYN-Based Focal Loss Approach for Credit Assessment". In: *Journal of Forecasting* (2025). DOI: 10.1002/for.3252.

[11] Kianeh Kandi and Antonio García-Dopico. "Enhancing Performance of Credit Card Model by Utilizing LSTM Networks and XGBoost Algorithms". In: *Machine Learning and Knowledge Extraction* 7.1 (2025), p. 20. DOI: 10.3390/make7010020.

[12] Saad M Darwish, Amr Ibrahim Salama, and Adel A Elzoghabi. "Intelligent approach to detecting online fraudulent trading with solution for imbalanced data in fintech forensics". In: *Scientific Reports* 15.1 (2025), pp. 1–19. DOI: 10.1038/s41598-025-01223-8.

[13] Ya-Han Hu, Chih-Fong Tsai, and Pei-Ting Wang. "Combining multiple data resampling methods and classifier ensembles for better financial dis-

tress prediction: homogeneous and heterogeneous approaches". In: *Annals of Operations Research* (2025), pp. 1–22. DOI: 10.1007/s10479-025-06706-5.

[14] Ines Gasmi, Sana Neji, Salima Smiti, and Makram Soui. "Features Selection for Credit Risk Prediction Problem". In: *Information Systems Frontiers* (2025), pp. 1–20. DOI: 10.1007/s10796-024-10559-x.

[15] Shujie Zou, Zhiming Cai, Chiawei Chu, Zefeng Zhao, Haohao Cai, Ning Shen, and Jie Ren. "Variable Selection and Fusion Sampling of Unbalanced Data in Credit Risk Assessment". In: *Computational Economics* (2025), pp. 1–31. DOI: 10.1007/s10614-025-10943-y.

[16] Huanjing Wang, Qianxin Liang, John T Hancock, and Taghi M Khoshgoftaar. "Feature selection strategies: a comparative analysis of SHAP-value and importance-based methods". In: *Journal of Big Data* 11.1 (2024), p. 44. DOI: 10.1186/s40537âĂŠ024âĂŠ00905âĂŠw.

[17] Shuoyan Lin, Dandan Song, Boyi Cao, Xin Gu, and Jiazhan Li. "Credit risk assessment of automobile loans using machine learning-based SHapley Additive exPlanations approach". In: *Engineering Applications of Artificial Intelligence* 147 (2025), p. 110236. DOI: 10.1016/j.engappai.2025.110236.

[18] Yan Zhang, Lin Chen, and Yixiang Tian. "A Method for Evaluating the Interpretability of Machine Learning Models in Predicting Bond Default Risk Based on LIME and SHAP". In: *arXiv preprint arXiv:2502.19615* (2025). DOI: 10.48550/arXiv.2502.19615.

[19] Seongil Han and Haemin Jung. "NATE: Non-pArameTric approach for Explainable credit scoring on imbalanced class". In: *PloS one* 19.12 (2024), e0316454. DOI: 10.1371/journal.pone.0316454.

[20] Victor Chang, Qianwen Ariel Xu, Shola Habib Akinloye, Vladlena Benson, and Karl Hall. "Prediction of bank credit worthiness through credit risk analysis: an explainable machine learning study". In: *Annals of Operations Research* (2024), pp. 1–25. DOI: 10.1007/s10479-024-06134-x.

[21] Idowu Aruleba and Yanxia Sun. "Effective credit risk prediction using ensemble classifiers with model explanation". In: *IEEE Access* (2024). DOI: 10.1109/ACCESS.2024.3445308.

[22] Xinjian Zhang, Yuting Zhang, Xuefeng Liu, and Ruping Wang. "Blockchain-Based Intelligent Risk Management Decision Support System for Supply Chain Financing". In: *International Journal of Intelligent Information Technologies (IJIIT)* 21.1 (2025), pp. 1–24. DOI: 10.4018/IJIIT.369153.

[23] Meng Xia, Zhijie Wang, Xingyu Lan, Wanan Liu, and Jiawei Wu. "Confidence-driven under-sampling decision forest for imbalanced credit scoring". In: *Engineering Applications of Artificial Intelligence* 147 (2025), p. 110278. DOI: 10.1016/j.engappai.2025.110278.

[24] Chenxi Wu. "Credit risk unveiled: Decision trees triumph in comparative machine learning study". In: *Applied and Computational Engineering* 79 (2024), pp. 174–181. DOI: 10.54254/2755-2721/79/20241613.

[25] Yao Zou, Meng Xia, and Xingyu Lan. "Interpretable credit scoring based on an additive extreme gradient boosting". In: *Chaos, Solitons & Fractals* 194 (2025), p. 116216. DOI: 10.1016/j.chaos.2025.116216.

[26] MN Sowmiya, S Jaya Sri, S Deepshika, and G Hanushya Devi. "Credit Risk Analysis using Explainable Artificial Intelligence". In: *Journal of Soft Computing Paradigm* 6.3 (2024), pp. 272–283. DOI: 10.36548/jscp.2024.3.004.

[27] Vishnu Ravi, Vineet Kumar Srivastava, Maninder Pal Singh, Ravi Kumar Burila, Nikhil Kassetty, Padma Naresh Vardhineedi, Venkata Reddy Pasam, Nuzhat Noor Islam Prova, and Indrajit De. "Explainable AI (XAI) for Credit Scoring and Loan Approvals". In: *arXiv* (2025).

[28] Sahab Zandi, Kamesh Korangi, María Óskarsdóttir, Christophe Mues, and Cristián Bravo. "Attention-based dynamic multilayer graph neural networks for loan default prediction". In: *European Journal of Operational Research* 321.2 (2025), pp. 586–599. DOI: 10.1016/j.ejor.2024.09.025.

[29] Xinyu Sun, Jiayu Liu, and Yan Zhang. "Enhancing Credit Risk Prediction through an Ensemble of Explainable Model". In: *Journal of Systems Science and Systems Engineering* (2025), pp. 1–22. DOI: 10.1007/s11518-025-5663-y.

[30] Scott M Lundberg and Su-In Lee. "A unified approach to interpreting model predictions". In: *Advances in neural information processing systems* 30 (2017).

[31] Marco Tulio Ribeiro, Sameer Singh, and Carlos Guestrin. "" Why should i trust you?" Explaining the predictions of any classifier". In: *Proceedings of the 22nd ACM SIGKDD international conference on knowledge discovery and data mining*. 2016, pp. 1135–1144. DOI: 10.1145/2939672.2939778.

[32] Rubén García-Céspedes, Francisco J Alias-Carrascosa, and Manuel Moreno. "On Machine Learning models explainability in the banking sector: the case of SHAP". In: *Journal of the Operational Research Society* (2025), pp. 1–13. DOI: 10.1080/01605682.2025.2485263.

[33] Nazerke Baisholan, J Eric Dietz, Sergiy Gnatyuk, Mussa Turdalyuly, Eric T Matson, and Karlygash Baisholanova. "FraudX AI: An Interpretable Machine Learning Framework for Credit Card Fraud Detection on Imbalanced Datasets". In: *Computers* 14.4 (2025), p. 120. DOI: 10.3390/computers14040120.

[34] Andres Alonso-Robisco and José Manuel Carbó. "Should We Trust the Credit Decisions Provided by Machine Learning Models?" In: *Computational Economics* (2025), pp. 1–30. DOI: 10.1007/s10614-025-10855-x.

[35] Thanh Thuy Do, Golnoosh Babaei, and Paolo Pagnottoni. "Explainable machine learning for credit risk management when features are dependent". In: *Measurement: Interdisciplinary Research and Perspectives* 22.4 (2024), pp. 315–340. DOI: 10.1080/15366367.2023.2261186.

[36] Lu Bai and Xuezhou Wen. "Improved Artificial Bee Colony Algorithm for Feature Selection to Enhance the Prediction of Credit Risk in SMEs". In:

Computational Economics (2025), pp. 1–34. DOI: 10.1007/s10614-025-10955-8.

[37] Murat Doğan, Muhammed Aslam Chelery Komath, and Özlem Sayilir. "Credit rating prediction with ESG data using data mining methods". In: *Future Business Journal* 11.1 (2025), p. 79. DOI: 10.1186/s43093-025-00490-1.

[38] Idowu Aruleba and Yanxia Sun. "An improved Ensemble Method with Data Resampling for Credit Risk Prediction". In: *IEEE Access* (2025). DOI: 10.1109/ACCESS.2025.3563432.

[39] Gang Yao, Xiaojian Hu, Pingfan Song, and Yue Zhang. "A novel class-imbalance-oriented feature selection method based on BPNNs and AdaBoost for enterprise credit risk prediction in the supply chain context". In: *International Journal of Information Technology & Decision Making* (2025). DOI: 10.1142/S0219622025500403.

[40] Ruilin Liang, Yuke Hou, Xin Liang, Bo Li, and XiaoXuan Yao. "The Application of Data Mining Models in Personal Credit Risk Control". In: *International Journal of Image and Graphics* (2025), p. 2750034. DOI: 10.1142/S0219467827500343.

[41] Sichong Lu, Xiaoming Zhang, Yi Su, Xiaojun Liu, and Lean Yu. "Efficient multimodal learning for corporate credit risk prediction with an extended deep belief network". In: *Annals of Operations Research* (2025), pp. 1–38. DOI: 10.1007/s10479-025-06612-w.

[42] Li Zeng and Wee-Yeap Lau. "Nexus Between ESG Performance and Credit Risk in Chinese FinTech Companies". In: *Asia-Pacific Financial Markets* (2025), pp. 1–29. DOI: 10.1007/s10690-025-09528-4.

[43] Guanglan Zhou and Shiru Wang. "Enhancing Credit Risk Decision-Making in Supply Chain Finance with Interpretable Machine Learning Model". In: *IEEE Access* 13 (2025), pp. 14239–14251. DOI: 10.1109/ACCESS.2025.3530433.

[44] Nazário Augusto de Oliveira and Leonardo Fernando Cruz Basso. "Advancing Credit Rating Prediction: The Role of Machine Learning in Corporate Credit Rating Assessment". In: *Risks* (2025). DOI: 10.3390/risks13060116.

[45] Wangke Lin and Yue Liu. "Deep Learning-Based Attention Mechanism Algorithm for Blockchain Credit Default Prediction". In: *International Journal of Advanced Computer Science & Applications* 16.2 (2025).

[46] P Sundaravadivel, R Augustian Isaac, D Elangovan, D KrishnaRaj, VV Rahul, and R Raja. "Optimizing credit card fraud detection with random forests and SMOTE". In: *Scientific Reports* 15.1 (2025), pp. 1–12. DOI: 10.1038/s41598-025-00873-y.

[47] Kianeh Kandi and Antonio Garcia Dopico. "Evaluating the Performance of Deep Convolutional Neural Networks and Support Vector Regression for Creditworthiness Prediction in the Financial Sector". In: *Inteligencia Artificial* 28.76 (2025), pp. 66–84. DOI: 10.4114/intartif.vol28iss76pp66-84.

[48] Lu Wang, Zecheng Yu, Jingling Ma, Xiaofang Chen, and Chong Wu. "A Two-Stage Interpretable Model to Explain Classifier in Credit Risk Prediction". In: *Journal of Forecasting* (2025). DOI: 10.1002/for.3288.

[49] Gustavo EAPA Batista, Ronaldo C Prati, and Maria Carolina Monard. "A study of the behavior of several methods for balancing machine learning training data". In: *ACM SIGKDD explorations newsletter* 6.1 (2004), pp. 20–29. DOI: 10.1145/1007730.100773.

[50] Nitesh V Chawla, Kevin W Bowyer, Lawrence O Hall, and W Philip Kegelmeyer. "SMOTE: synthetic minority over-sampling technique". In: *Journal of artificial intelligence research* 16 (2002), pp. 321–357. DOI: 10.1613/jair.953.

[51] Yao Zou and Guanghua Cheng. "Level-Wise Feature-Guided Cascading Ensembles for Credit Scoring". In: *Symmetry* 17.6 (2025), p. 914. DOI: 10.3390/sym17060914.

[52] Gustavo EAPA Batista, Ana LC Bazzan, Maria Carolina Monard, et al. "Balancing training data for automated annotation of keywords: a case study." In: *Wob* 3 (2003), pp. 10–18.

[53] Idowu Aruleba and Yanxia Sun. "Enhanced credit risk prediction using deep learning and SMOTE-ENN resampling". In: *Machine Learning with Applications* 21 (2025), p. 100692. DOI: 10.1016/j.mlwa.2025.100692.

[54] Nurafni Damanik and Chuan-Ming Liu. "Advanced Fraud Detection: Leveraging K-SMOTEENN and Stacking Ensemble to Tackle Data Imbalance and Extract Insights". In: *IEEE Access* (2025). DOI: 10.1109/ACCESS.2024.

[55] Leonardo Canete-Sifuentes, Raul Monroy, and Miguel Angel Medina-Perez. "FT4cip: A new functional tree for classification in class imbalance problems". In: *Knowledge-Based Systems* 252 (2022), p. 109294. DOI: 10.1016/j.knosys.2022.109294.

[56] Erwis Melchor-Pérez, José Fco. Martínez-Trinidad, J. Ariel Carrasco-Ochoa, and Agustin Santiago-Alvarado. "Oversampling and instance selection to predict credit risk". In: *International Journal of Applied Pattern Recognition* (2025). in press. DOI: 10.1504/IJAPR.2025.10073009.

[57] Mimi Mukherjee and Matloob Khushi. "SMOTE-ENC: A novel SMOTE-based method to generate synthetic data for nominal and continuous features". In: *Applied system innovation* 4.1 (2021), p. 18. DOI: 10.3390/asi4010018.

[58] Z Huang. "Clustering Large Data Sets With Mixed Numeric And Categorical Values," Proceedings Of 1st Pacific-Asia Conference on Knowledge Discouvery And Data Mining". In: (1997).

[59] Zhexue Huang. "Extensions to the k-means algorithm for clustering large data sets with categorical values". In: *Data mining and knowledge discovery* 2.3 (1998), pp. 283–304. DOI: 10.1023/A:1009769707641.

[60] Duy-Tai Dinh, Tsutomu Fujinami, and Van-Nam Huynh. "Estimating the optimal number of clusters in categorical data clustering by silhouette coef-

ficient". In: *International Symposium on Knowledge and Systems Sciences*. Springer. 2019, pp. 1–17. DOI: 10.1007/978-981-15-1209-4_1.

[61] Dendy Arizki Kuswardana, Dwi Arman Prasetya, Trimono Trimono, and I Gede Susrama Mas Diyasa. "Comparison of Elbow and Silhouette Methods in Optimizing K-Prototype Clustering for Customer Transactions". In: *Jurnal Ilmiah Edutic: Pendidikan dan Informatika* 12.1 (2025), pp. 43–48. DOI: 10.21107/edutic.v12i1.29744.

[62] Milton García-Borroto, José Fco Martínez-Trinidad, Jesús Ariel Carrasco-Ochoa, Miguel Angel Medina-Pérez, and José Ruiz-Shulcloper. "LCMine: An efficient algorithm for mining discriminative regularities and its application in supervised classification". In: *Pattern Recognition* 43.9 (2010), pp. 3025–3034. DOI: 10.1016/j.patcog.2010.04.008.

[63] Octavio Loyola-González, Miguel Angel Medina-Pérez, José Fco Martínez-Trinidad, Jesús Ariel Carrasco-Ochoa, Raúl Monroy, and Milton García-Borroto. "PBC4cip: A new contrast pattern-based classifier for class imbalance problems". In: *Knowledge-Based Systems* 115 (2017), pp. 100–109. DOI: 10.1016/j.knosys.2016.10.018.

[64] Idowu Aruleba and Yanxia Sun. "An improved ensemble method with data resampling for credit risk prediction". In: *IEEE Access* (2025). DOI: 10.1109/ACCESS.2025.3563432.

[65] Fatma M Talaat, Abdussalam Aljadani, Mahmoud Badawy, and Mostafa Elhosseini. "Toward interpretable credit scoring: integrating explainable artificial intelligence with deep learning for credit card default prediction". In: *Neural Computing and Applications* 36.9 (2024), pp. 4847–4865. DOI: 10.1007/s00521-023-09232-2.

[66] Chenlu Zheng, Futian Weng, Yiwen Luo, and Cai Yang. "Micro and small enterprises default risk portrait: evidence from explainable machine learning method". In: *Journal of Ambient Intelligence and Humanized Computing* 15.1 (2024), pp. 661–671. DOI: 10.1007/s12652-023-04722-6.